

- $\text{Var}(X) = E[X^2] - E[X]^2$.
- $\text{Cov}(X, Y) := \mu_{XY} - \mu_X \mu_Y = E[(X - \mu_X)(Y - \mu_Y)]$
- **Markov**: if $X \geq 0$, then $P(X \geq a) \leq \frac{E[X]}{a} \forall a > 0$.
- **Chebyshev**: if $\sigma^2 < \infty$, $P(g(X) > a^2 \sigma^2) \leq \frac{E[g(X)]}{a^2 \sigma^2}$.
- Total probability: if $\{B_i : i \in I\}$ is a partition of the sample space, then

$$P(A) = \sum_{i \in I} P(A | B_i) P(B_i).$$

- Total expectation: $E[X] = E[E[X | Y]]$.
- Total var: $\text{Var}(X) = E[\text{Var}(X | Y)] + \text{Var}(E[X | Y])$.
- Bayes's Theorem: $P(A | B) = \frac{P(B|A)P(A)}{P(B)}$.
- Markov chain: future independent of the past, i.e., X_{n+1} is at most dependent on X_n .
- Transition probability: $p_{ij}^{n,m} = P(X_m = j | X_n = i)$.
- Transition probability matrix: if π_i is the distribution of X_i , then $\pi_t = \pi_0 \prod_{i=0}^{t-1} \mathbf{P}^{i,i+1}$.
- Stationary Markov chain: TPM is independent of the timestamp n .
- **Stochastic matrix**: non-negative matrix such that row sums are 1.
- **Chapman-Kolmogorov**: $\mathbf{P}^{(m)} = \mathbf{P} \mathbf{P}^{(m-1)} = \mathbf{P}^{(m-1)} \mathbf{P}$. If X is stationary, then for all $m, n \in \mathbb{N}$,

$$P_{ij}^{0,m+n} = \sum_{k \in S} P_{ik}^{0,m} P_{kj}^{0,n},$$

and $P(X_n = j | X_0 = i) = (\mathbf{P}^n)_{ij}$.

- if $\mu := [f(x_1) \ f(x_2) \ \cdots \ f(x_s)]^T$ for some function f on the state space, then

$$(\mathbf{P}^n \mu)_i = \sum_{j=1}^s P_{ij}^n \mu_j = \sum_{j=1}^s P(X_n = x_j | X_0 = x_i) f(x_j) = E[f(X_n) | X_0 = x_i].$$

Suppose $X_0 \sim \lambda$, then clearly

$$E[f(X_n)] = \sum_{i=1}^s E[f(X_n) | X_0 = x_i] \lambda_i = \sum_{i=1}^s \lambda_i (\mathbf{P}^n \mu)_i = \lambda \mathbf{P}^n \mu.$$

- **Stationary distribution**: $\pi = \pi \mathbf{P}$.
- If λ is an eigenvalue of $\mathbf{P}$, then $|\lambda| \leq 1$.
- **Absorbing state**: for all $j \neq i$, we have $P_{ij} = 0$.
- $x \leftrightarrow y \iff \exists m, n \in \mathbb{N}$ such that $P_{xy}^{(m)}, P_{yx}^{(n)} > 0$.
- **Irreducible chain**: only one class.

- **Return prob.**: $P_{ii}^{(n)} = P(X_n = i | X_0 = i)$.
- **1st return prob.**: $f_{ii}^{(n)} = P(X_j \neq i \forall j < n, X_n = i | X_0 = i)$. $f_{ii}^{(0)} = 0$, $f_{ii}^{(n)} \leq P_{ii}^{(n)}$.
- $P_{ii}^{(n)} = \sum_{k=0}^n f_{ii}^{(k)} P_{ii}^{(n-k)}$.
- $f_{ii} = \sum_{n=0}^{\infty} f_{ii}^{(n)}$ is the prob. of returning to i in finite time. i is **recurrent** if $f_{ii} = 1$ and **transient** if $f_{ii} < 1$.
- Recurrent: $P(\sum_{n=1}^{\infty} I\{X_n = i\} = \infty | X_0 = i) = 1$ so

$$E\left[\sum_{n=0}^{\infty} I\{X_n = x\} | X_0 = x\right] = \sum_{n=0}^{\infty} \mathbf{P}^n(x, x) = \infty,$$

but $\mathbf{P}^n(x, x)$ may converge to 0.

- Number of revisits to i : $N_i \sim \text{Geo}(1 - f_{ii})$. $E[N_i | X_0 = i] = \frac{f_{ii}}{1 - f_{ii}}$ and expected number of visits including the initial one is $\frac{1}{1 - f_{ii}}$.
- i is a transient state iff $\sum_{n=1}^{\infty} P_{ii}^{(n)}$ is finite.
- If i is transient, then $\lim_{m \rightarrow \infty} \sum_{n=m}^{\infty} P_{ii}^{(n)} = 0$ by monotone convergence theorem.
- Irreducible recurrent: $P(\sum_{n=0}^{\infty} I\{X_n = i\} = \infty) = 1$
- Irreducible transient: $P(\sum_{n=0}^{\infty} I\{X_n = i\} < \infty) = 1$
- All finite-state irreducible chains are recurrent.
- Reducible chains will enter one of the recurrent classes in the long-run.
- **Period**: $d(i) := \gcd\{n \in \mathbb{N}^+ : P_{ii}^{(n)} > 0\}$. i is **aperiodic** iff $d(i) = 1$.
- If $i \leftrightarrow j$, then $d(i) = d(j)$.
- $\forall i \in S, \exists N \in \mathbb{N}$ s.t. $\forall n \geq N, P_{ii}^{(n-d(i))} > 0$ and $P_{ji}^{(m+n-d(i))} > 0$ whenever $P_{ji}^{(m)} > 0$.
- If $\mathbf{P}$ is the TPM for a finite-state irreducible aperiodic chain, then $\exists N \in \mathbb{N}^+$ such that $\mathbf{P}^{(N)}$ has all positive entries (definition for regular chain).
- Irreducible, aperiodic, finite-state $\implies$ regular chain.
- Regular $\implies$ irreducible.
- $\mathbf{P}^{(k)}$ is regular $\implies \mathbf{P}^{(n)}$ is regular $\forall n \geq k$.
- Let $\mathbf{P}$ be a regular TPM for some regular Markov chain with finite state space, then there is a unique stationary distribution π with $\pi_j = \lim_{n \rightarrow \infty} P_{ij}^{(n)}$ for all i, j . $\pi_j = P(X_n = j)$ is the proportion of time where $X_n = j$ in the long-run.
- Local bal.: $\pi_i P_{ij} = \pi_j P_{ji}$ for all $i, j \implies$ global bal..
- **Stopping time**: $T_A := \min\{n \in \mathbb{N} : X_n \in A\}$ is the first time X enters A .
- **Null recurrent**: $\mathbb{E}[T_i | X_0 = i] = \infty$. **Positive recurrent**: $\mathbb{E}[T_i | X_0 = i] < \infty$.
- **Ergodic**: irreducible aperiodic positive recurrent.
- Define $m_i := \mathbb{E}[R_i | X_0 = i] = \sum_{n=1}^{\infty} n f_{ii}^{(n)}$ in ergodic chains, then $\lim_{n \rightarrow \infty} P_{ij}^{(n)} = \lim_{n \rightarrow \infty} P_{jj}^{(n)} = \frac{1}{m_j} = \pi_j$ gives a stationary distribution.

- Null recurrent: $\lim_{n \rightarrow \infty} P_{ij}^{(n)} = 0$.
- Irreducible recur.: $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n P_{ij}^{(k)} = \frac{1}{m_j}$. If aperiodic: $\lim_{n \rightarrow \infty} P_{ij}^{(n)} = \frac{1}{m_j}$; else $\lim_{n \rightarrow \infty} P_{jj}^{(nd)} = \frac{d}{m_j}$.
- If $f(x) = P(T_A < T_B \mid X_0 = x)$, then for all $x \notin A \cup B$,

$$\begin{aligned} f(x) &= \sum_{y \in S} P(T_A < T_B \mid X_1 = y, X_0 = x) P(X_1 = y \mid X_0 = x) \\ &= \sum_{y \in S} P(T_A < T_B \mid X_0 = y) P(X_1 = y \mid X_0 = x) \\ &= \sum_{y \in S} P_{xy} f(y). \end{aligned}$$

• **First-step analysis:**

1. Identify quantity of interest $a_i(T) = h(i, X_1, \dots, X_T \mid X_0 = i)$.
 2. $a_i(T) = \sum_{k \in S} h(\cdot \mid X_1 = k, X_0 = i) P(X_1 = k \mid X_0 = i)$.
 3. Consider $Y_n = X_{n+1}$ and establish $h(\cdot \mid X_1 = k, X_0 = i) = g_i(a_k(T))$.
 4. Solve the system.
- Gambler's ruin: $X_0 = k$ for $0 < k < N$ with winning probability p .
 - Fair game: $P(X_T = 0 \mid X_0 = k) = 1 - \frac{k}{N}$ and $E[T \mid X_0 = k] = k(N - k)$.
 - Otherwise:

$$\begin{aligned} P(X_T = 0 \mid X_0 = k) &= 1 - \frac{1 - (q/p)^k}{1 - (q/p)^N}, \\ E[T \mid X_0 = k] &= \frac{1}{p - q} \left[\frac{N(1 - (q/p)^k)}{1 - (q/p)^N} - k \right] \end{aligned}$$

- Random walk: $\frac{\xi_{i+1}}{2} \sim \text{Bernoulli}(p)$, $\frac{X_{n+1}}{2} \mid X_0 = 0 \sim \text{Bin}(n, p)$.
- Hastings-Metropolis:
 - p is target distribution with sample space S and M is the max depth of iterations.
 - Fix an irreducible chain with state space S and TPM Q which is symmetric. Select some $X_0 \in S$.
 - At the n -th iteration:
 - * Generate $Y \sim q$ s.t. $q_j = Q_{X_n j}$ for all $j \in S$ and $U \sim U(0, 1)$.
 - * Compute $\alpha(X_n, Y) := \min \left\{ \frac{p_Y Q_{Y X_n}}{p_{X_n} Q_{X_n Y}}, 1 \right\}$.
 - * $U < \alpha(X_n, Y) ? X_{n+1} = Y : X_{n+1} = X_n$.
 - Return $Z := X_M \sim p$ approximately.

- $P(Z_{n+1} = j \mid Z_n = i) = Q_{ij} \alpha(i, j) = P_{ij}$
- $P(Z_{n+1} = i \mid Z_n = i) = Q_{ii} + \sum_{k \neq i} Q_{ik} (1 - \alpha_{ik}) = P_{ii}$
- If $p = c\mathbf{b}$ where $\mathbf{b}$ is a kernel function, only need $\frac{b_i}{b_j}$.
- Prob. gen. func. $\phi_X(t) := \mathbb{E}[t^X] = \sum_k P(X = k) t^k$.
- $\phi_X^{(n)}(t) = \sum_{k=n}^{\infty} k! P(X = k) t^{k-n}$, so $P(X = x) = \frac{\phi_X^{(x)}(0)}{x!}$
- $\phi_{X+Y} = \phi_X \phi_Y$ if independent.
- $\phi_{X_{n+1} \mid X_n = x}(t) = \mathbb{E}[t^{X_{n+1}} \mid X_n = x] = \mathbb{E}[\prod_{i=1}^x t^{\xi_i}] = \prod_{i=1}^x \phi_{\xi_i}(t)$
- $\phi_{X_n \mid X_0 = k}(t) = \phi_{X_1 \mid X_0 = k}(\phi_{\xi}^{n-1}(t))$, so $\phi_{X_n \mid X_0}(t) = \phi_{\xi}^n(t)^{X_0}$
- $\mathbb{E}[X_n \mid X_0] = \mu^n X_0$ and $\text{Var}(X_n \mid X_0) = \begin{cases} \frac{1-\mu^n}{1-\mu} (\mu^{n-1} \sigma^2) X_0 & \text{if } \mu \neq 1 \\ n (\mu^{n-1} \sigma^2) X_0 & \text{if } \mu = 1 \end{cases}$.
- **Extinction Prob.:** $u_n^{(k)} := P(X_n = 0 \mid X_0 = k) = u_n^k$
- $u_n = \sum_{i \in S} P(X_1 = i \mid X_0 = 1) u_{n-1}^{(i)} = \sum_{i \in S} P(\xi = i) u_{n-1}^i$
- $u_n^{(k)} = \phi_{\xi}^{n-1}(P(\xi = 0))^k$
 - if $P(\xi = 0) = 0$, then $u_{\infty}^{(k)} = 1$;
 - if $P(\xi = 0) > 0$, then if $\mathbb{E}[\xi] \leq 1$, $u_{\infty}^{(k)} = 1$; else, $u_{\infty} \in [0, 1)$ is s.t. $u_{\infty} = \phi_{\xi}(u_{\infty})$
- Pois. Proc.: $X(0) = 0$, $X(s+t) - X(s) \sim \text{Pois}(\lambda t)$ and for any $0 = t_0 < t_1 < \dots < t_n$, $X(t_i) - X(t_{i-1})$'s are mutually independent
- Pois. Pt. Proc.:
 - for any $0 = t_0 < t_1 < \dots < t_n$, $N((t_i, t_{i+1}])$'s are mutually independent;
 - $P(N((t, t+h]) \geq 1) = \lambda h + o(h)$ and $P(N((t, t+h]) \geq 2) = o(h)$ as $h \rightarrow 0$.
- Waiting time: $W_n \sim \text{Gamma}(n, \lambda)$ s.t. $f_{W_n}(t) = \frac{\lambda^n t^{n-1}}{(n-1)!} e^{-\lambda t}$.
- Sojourn time: $S_n = W_1 \sim \text{Exp}(\lambda)$ where $f_{S_n}(t) = \lambda e^{-\lambda t}$
- $W_1 \mid X(t) = 1 \sim U(0, t)$. Joint is the sum of independent uniform r.v.

$$\begin{aligned} f(w_1, w_2, \dots, w_n \mid X(t) = n) &= \frac{P(S_n > t - w_n) \prod_{i=1}^n f_{S_{i-1}}(w_i - w_{i-1})}{\frac{(\lambda t)^n e^{-\lambda t}}{n!}} \\ &= \frac{n! e^{-\lambda(t-w_n)} \prod_{i=1}^n \lambda e^{-\lambda(w_i - w_{i-1})}}{(\lambda t)^n e^{-\lambda t}} = \frac{n!}{t^n}. \end{aligned}$$

$$f_{W_k \mid X(t)=n}(x) = \frac{n! x^{k-1} (t-x)^{n-k}}{(n-k)!(k-1)! t^n}$$